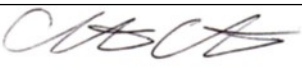


Guideline Title: Blood Product Administration

Guideline Number: GUID-3203-PR002

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To provide safe and effective blood and blood product administration from a sending facility and to recognize and effectively treat transfusion reactions.
- II. **DEFINITIONS:** None
- III. **DEPARTMENT GUIDELINE:**
 - a. **EQUIPMENT:**
 - i. Personal protective equipment.
 - ii. Appropriate blood product.
 - iii. Filtered blood tubing or leukocyte reduction filter.
 - iv. A one-liter bag of 0.9 % Normal Saline (NS).
 - v. Pressure bag
 - b. **INDICATIONS:**
 - i. Hypovolemia with acute blood loss.
 - ii. Restoration of clotting factors with FFP
 - c. **CONTRAINDICATIONS:**
 - i. Patient's refusal of blood or blood products.
 - ii. The unavailability of type specific blood or Type O negative
 - d. **PROCEDURE:**
 - i. Utilize appropriate protective equipment.
 - ii. Initially assess the following:
 1. Vital signs.
 2. Skin condition; i.e. rashes, redness, etc.
 3. Size and integrity of IV site.
 - a. Adults: Prefer 18 g for PRBCs and 20 g for other products.
 - b. Pediatrics: 22-24 g.
 - c. Never transfuse blood in piggyback fashion into any IV with medications infusing other than NS.
 4. Breath sounds.

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5. Fluid volume status.
6. History of previous transfusion reactions.
- iii. If available, obtain copy of patient's blood type & screen from transferring facility.
- iv. Verify order for blood product transfusion. Ensure consent form has been signed.
- v. Ideally, the blood should be warm.
- vi. Check the unit of blood for: (have two providers check simultaneously)
 1. Patient's name.
 2. Patient's blood type.
 3. Blood product type.
 4. Blood product ABO type.
 5. Unit number.
 6. Expiration date.
- vii. Confirm that the above are correct and consistent with the patient.
- viii. Once the unit of blood and the patient have been correctly correlated, document the person hanging the blood. Sign with date and time.
- ix. Flush IV with 0.9 % NS until clear.
- x. May place blood in pressure bag and inflate the bag not to exceed 300 mmHg.
- xi. Start transfusion slowly; assess vital signs every 10 minutes. Watch for a transfusion reaction or complications.
- xii. If a transfusion reaction occurs:
 1. Stop transfusion immediately.
 2. Assess vital signs.
 3. Remove blood bag and tubing and set aside.
 4. Assure bag and tubing are given to receiving RN/appropriate medical staff.
 5. Provide systemic support as required.
 6. Transfusion reactions can be defined as shown in the chart below.
- xiii. When transfusion is complete:
 1. Flush IV with 0.9 % NS until clear
 2. Record time of completion on tag
- xiv. Additional Points:
 1. Pediatric transfusion is 10 ml/kg.
 2. All transfusions should be completed within 4 hours.
 3. Review potassium and calcium levels if multiple transfusions received.

IV. DOCUMENTATION:

- a. EMS Charts - Unit #, Blood product type (FFP, PRBC, factor etc.), Time transfusion was initiated and completed, Volume and rate infused, Reaction yes/no, Vital signs before transfusion and when completed

V. REFERENCES: None

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TRANSFUSION REACTIONS				
Type	Onset	Cause	S/S	Treatment
Febrile	First 15 mins	Presence of bacterial pathogens Sensitivity to leukocytes / platelets Hemolytic episode	Fever and shaking Chills within 30 min. Shock soon after	NS infusion Tylenol Systemic support Vasopressor infusion per GUID-3203-M011
Allergic	15-30 mins	Sensitivity to plasma protein	Hives Generalized itching Wheezing Anaphylaxis	Benadryl 25-50mg IVP - If patient has history of reactions pre-medicate with Benadryl
Septic	15-30 mins	Contaminated blood with bacteria	Chills High Fever Vomiting & diarrhea Hypotension	NS infusion Vasopressor infusion per GUID-3203-M011 Solumedrol 125mg IVP
Hemolytic	30 mins	Blood incompatible with patient Leading to antibodies in patient's plasma rapidly combining with donor RBCs and cells are hemolyzed	Chills Low Back Pain HA Nausea CP Fever Hypotension Vascular collapse	NS infusion Vasopressor infusion per GUID-3203-M011